

Identification and Detection of Toxic Comments using Supervised Classification

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Abstract— In an era of ubiquitous digital communication, the proliferation of toxic comments on online platforms presents enormous problems to preserving meaningful conversations. This study presents an approach to identify toxic comments using supervised classification techniques. The dataset of Wikipedia talk pages is used, presented by Google Jigsaw. The methodology involves text preprocessing, feature extraction using GloVe and training multiple classifiers including Support Vector Machine, Logistic Regression, XGBoost, Convolutional Neural Network and Bi-directional Long Short-term memory. The results demonstrate that Bi-directional LSTM achieved test accuracy of 92.06% and F1-score 89.8, outperforming other classifiers. Furthermore, this study provides more insights into classification performance using other performance metrics including probability calibration curve and multi label confusion matrix. This project adds to the work targeted at combating online toxicity and gives realistic foundation for automated moderation systems, hoping to build more inclusive online platforms.

Keyword-- Long-Short Term Memory, Convolutional Neural Network, Support Vector Machine, Porter Stemmer, GloVe, Logistic Regression, XGBoost.

I. INTRODUCTION

Online platforms, once praised as bastions of free expression, now-a-days are grappling with the persistent issue of toxic comments, emphasizing the need for robust classification systems. In the context of online discussions, "toxicity" refers to statements that are insulting or destructive. These remarks, which frequently contain hate speech, threats, or harassment, not only degrade the quality of online discourse, but also have major consequences for individuals and communities. The abundance of toxic comments on social media platforms and news articles has become a major issue in the digital age. Therefore, classification of toxic comments hold significant importance in various domains.

Classification of toxic comments raises a number of issues. To begin with, the subjective and context-

dependent characteristic of toxicity makes it difficult to develop concrete definitions. What is considered toxic in one context could be tolerable in another. Furthermore, datasets used to train models are often imbalanced, with toxic comments being a small fraction, resulting in biased classifiers. Moreover, the dynamic nature of language and the introduction of novel types of toxicity necessitate adaptive and ever-changing classification methods [1].

Several projects and research have been conducted to address toxic comments classification challenges. Choosing the most effective technique among others is critical to successful classification. While prior research studies have made substantial progress, there is still a need to address issues such as model bias, adaptation to new types of toxicity, and efficiency in large-scale applications. This project aims to advance the subject of toxic comment classification by assessing and comparing previous research, finding strengths and drawbacks, and suggesting an approach effective in capturing toxic comments.

The main objective of this project is to improve the model's ability to discern between toxic and non-toxic comments by incorporating machine learning techniques such as deep learning and ensemble methods. Effective feature extraction is required to capture the subtle properties of toxic language. We used sampling and feature extraction approaches such as word embeddings (GloVe), stemming, and stop-words removal to extract meaningful features from data. In addition, we explored with different model architectures, such as recurrent neural networks (RNNs), convolutional neural networks (CNNs) and base machine learning classifiers, to find the best configuration for toxic comment classification.

II. LITERATURE REVIEW

Toxic comments classification has emerged as an active research area in last few years, featuring ever evolving challenges, newly proposed methodologies, and approaches.

Nobta et al. lists several challenges in comments classification. The first challenge with simple classification is that users can fool simple detection techniques. Another challenge is the employment of stylistic elements in online debates, such as irony to indicate sarcasm or referencing potentially harmful information. It is also uncertain which categorization strategy to employ and how to identify classes accurately from one another. As a result of this ambiguity, researchers have developed several annotation rules, and the resulting datasets utilize diverse labels.

Julian Risch et al. states switching to more fine-grained labeling, encounter new problems. Firstly, less accessible training data per class. Secondly, increased complexity annotating with fine-grained classification, the number of accessible samples per class decreases. It is a significant difficulty to gather enough samples per class without adding a bias into the sampling from a basic population of comments. The class imbalance hinders neural network training, necessitating the implementation of remedies.

Sampling changes the dataset such that there are an equal number of samples from each class. To accomplish this, redundant samples from the majority class can be removed, or samples from the minority class can be sampled again. Another option is to utilize a weighted loss function, which modifies the training process by increasing penalties for mistakes in the minority class compared to the majority class. Another approach is the synthetic minority over-sampling technique (SMOTE), which has already been used to improve a dataset of harsh remarks [3].

The selection of word embedding, and deep learning models is critical for better results. Word embeddings are n-dimensional distributed representations of a text that try to capture the meanings of the words. Word embeddings learned using Word2Vec are more effective at capturing word semantics and leveraging word relatedness. Word2Vec focuses entirely on information acquired from the local context window, ignoring global statistic data. The GloVe is an LSA-CBOW hybrid that is efficient and scalable for big corpora. The GloVe is a common model based on the global co-occurrence matrix, with each element in the matrix representing the frequency with which the words co-occur in a certain context frame. [5].

The development in deep learning algorithms give compelling results for text classification, particularly the use of Convolutional Neural Networks (CNN) that emphasize in the structure of words in the documents. **Spiros et al.** employed this learning technique on a

large dataset to identify toxic comments. To prove CNN enhanced toxic comments classification, they compared their findings against bag-of- words approach combined with selection of algorithms.

Other proposed solutions include bi-directional recurrent neural networks with attention (**Pavlopoulos et al.**). However, some classifiers deal with task's challenges, other challenges remain unresolved, requiring additional research (**Betty van Aken et al.**). For this purpose, they compared different models and shallow approaches on a large dataset, to find the model with better performance and result, to carry out in-depth error analysis.

For analysis, F1 measure is preferred as it is more indicative towards strongly unbalanced datasets. To calculate the F1 measure, binary predictions are required. To convert the continuous output for the multi-label task into binary labels, estimates of appropriate threshold values per class are used. The probability calibration curve provides a measure of classifier performance without the need for a specific threshold, which is added to provide additional comparability of the results [6].

III. RESEARCH METHODOLOGY

This section introduces the process of feature extraction and classification of textual data for identification of toxic comments. Figure 1 shows the schematic diagram of the workflow used in this project. The dataset used in this project is Wikipedia talk pages comments presented by Google Jigsaw on Kaggle comprising more than 150000 comments. Data contains multi-class labels. The only feature of dataset is online comments, that are classified into following classes: obscene, threat, identity hate, insult, toxic, severe toxic.

1. Text pre-processing

1.1. Tokenization

To prepare textual data for analysis, firstly, we checked null values in dataset. Then natural language tool-kit's tokenizer is applied to tokenize sentences into words.

1.2. Word Stemming

Stemming is applied on tokenized words reduce word to its base form. It normalizes the text and improves the vocabulary size.

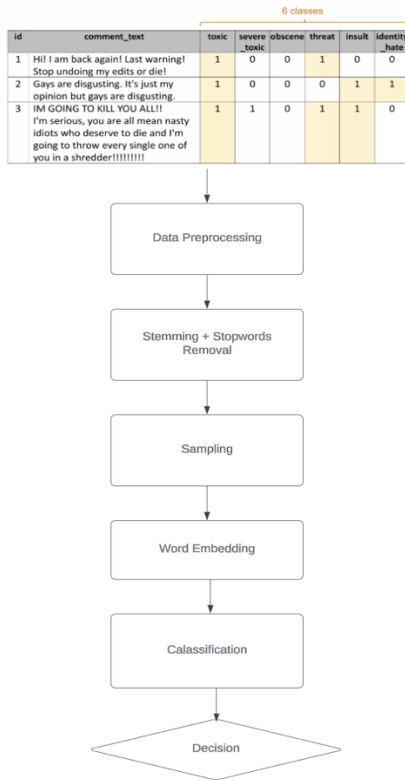


Figure 1: Outline of key steps in building models and classification of comments.

For this purpose, Porter Stemmer algorithm is used, a rule-based metaheuristic approach removes common morphological and inflectional endings from words. Figure 2 shows the schematic diagram of the rules for Porter Stemmer Algorithm.

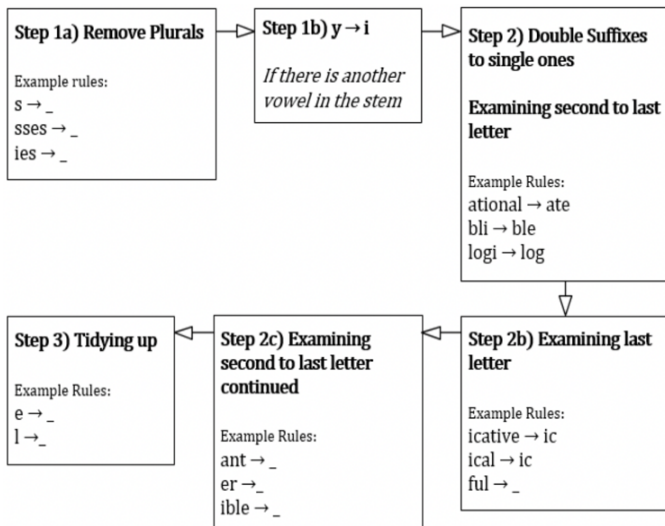


Figure 2: Overview: Steps of rules for Porter Stemmer Algorithm.

1.3. Removal of Stop-words

The next pre-processing technique used is Stop words Removal. The focus of this technique is to filter out common words (i.e. this, is, it, an etc.) to focus on more meaning full words. Natural Language Tool Kit in python is used, contains prebuilt lists of stop words for various languages. It is case in-sensitive pre-processing technique, checks words in stop words list. If word is matched, it removes both upper- and lower-case versions of stop words.

1.4. Word Embedding techniques

Word embeddings are dense vector representations of words in a continuous vector space, allowing word to capture syntactic and semantic relationship between the words. There are various types of embedding techniques. One is Pre-trained Word Embedding including GloVe, Word2Vec and other is Contextual Word Embedding including BERT. The GloVe is the technique used in this project.

1.4.1. Global Vector for Word Representation

GloVe is primarily a log-bilinear model with a weighted least-squares aim. The model's fundamental concept is based on the simple observation that word-word co-occurrence probability ratios have the potential to encode some type of meaning. [4]

To convert textual data to their respective vectors, 50-dimensional word vector is used. Following steps are followed by GloVE:

1. Covert all pre-processed words to lower case to ensure uniformity in structure.
2. Create a vocabulary that maps each unique word to index.
3. Initialize word embedding for each word in the vocabulary by using pre-trained embedding.
4. Replace each word in text with its corresponding word vector.

1.5. Sampling

Sampling is crucial step in machine learning pipeline, selecting subset of examples from large dataset, helps in managing computational resources, reducing training time and improving generalization performance of model. Certain classes in dataset are imbalanced that can lead model to bias towards some classes.

To sampling dataset, stratified sampling is used ensuring that distribution in each class should be representative of whole dataset to avoid bias. Stratified

sampling selects data in-order that can lead to underfitting. To mitigate this issue, random sampling is used to select data randomly increases variation in data. The 75/25 train-test split used in this project.

1.6. Classification

To classify toxic comments, this project used variety of classification algorithms such as supervised learning classifiers, ensemble technique and deep models to find the best among others.

1.6.1. Bi-directional Long Short-Term Memory

Long Short-term memory is type of recurrent neural network solve the problem of vanishing gradient occur in traditional recurrent neural network. It contains gates control flow of information in and out of memory cells that remember the long-term dependencies of long sequences. Three epochs were used by keeping in view computational consumption of time and storage. The architecture used in this project has following layers:

- a. Embedding layer: Convert input sequences into dense vector of fixed size. (embedding_size = 128)
- b. Bi-directional LSTM layer: Processed input sequences both in forward and backward direction, update hidden state and concatenate output of each direction at each timestamp. (lstm_units = 64)
- c. GlobalMaxPooling1D layer: Aggregates the output of the LSTM layer across the entire sequence by taking the maximum value for each feature.
- d. Dense Layer: The first dense layer has 50 units with ReLU activation function, mapping previous layer output to next layer and introduce non-linearity.
- e. Drop-out Layer: Regularization technique to prevent overfitting. (dropout = 0.1).
- f. Dense layer: The second dense layer has 6 units with sigmoid activation for multilabel classification.

1.6.2. Convolutional Neural Network

For convolutional neural network, the previous neural network architecture is used contain Convolutional Layer instead of Bi-directional LSTM layer. For convolution layer, 1D Convolutional layer with 64 num_filters and (kernel_size = 3) filter size. ReLU is used as activation function of this layer.

For optimization, Adam optimizer and binary cross entropy function is used to calculate loss function in this project.

1.1.1. Logistic Regression

Logistic regression is commonly employed in binary classification tasks, where the outcome is either 0 or 1. When it comes to multi-label classification, where each instance can belong to multiple classes, logistic regression can be modified in a variety of ways. One common strategy is to utilize a technique called "One-vs-Rest".

The logistic function (also called the sigmoid function) is used for this purpose:

$$P(Y = 1 | X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \dots + \beta_n X_n)}}$$

The parameters used for this algorithm are newton-cg solver used as optimization algorithm, avoid model to overfit the data and converge faster for large datasets.

1.1.2. Support Vector Machine

SVM seeks to identify the best hyperplane that divides data points from distinct classes while maximizing the margin. The kernel used for support vector machine is Radial basis function, a common choice for Support Vector Machine models, particularly when dealing with data that is not linearly separable. It enables SVMs to describe complicated decision boundaries by converting input characteristics to a higher-dimensional space.

1.1.3. XGBoost

XGBoost is powerful ensemble technique used multiple weak learners to make a strong predictive model. The weak learners used are decision trees. The parameters selected are number of rounds that corrects prediction of previous tree set to 100, maximum depth set to 4 for early termination and avoid overfitting. Value of learning rate set to 0.1 which led model to more robust convergence.

IV. ANALYSIS AND RESULTS

For analysis, algorithms are compared with each other based on performance metrics including multi-label confusion matrix, probability calibration curve, F1 score and accuracy. Bi-directional Long Short-term memory performed well in classification and outperforms all other classifiers. Results for algorithms performance are detailed in Table below.

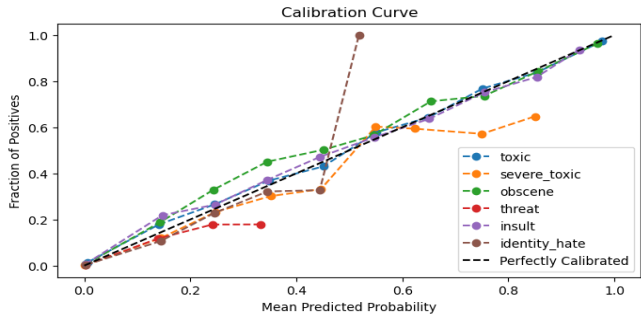
	LSTM	CNN	SVM	XGBoost	Logistic Regression
Accuracy	92.06	91.77	90.67	90.53	90.48
F1 score	89.8	89	72.9	75.9	73.7

Accuracy is not accurate metric to use when dealing with imbalanced data. So, F1-score is used as well to measure the performance of each model.

Probability Calibration Curve

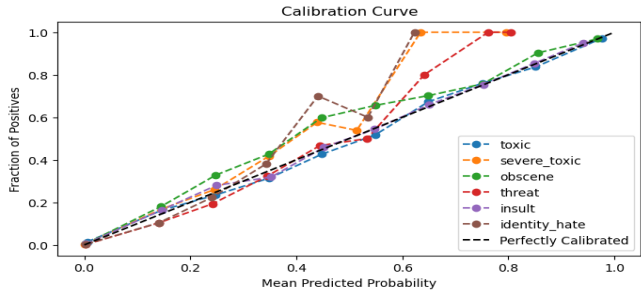
A probability calibration curve is a graphical depiction used to analyze how well the projected probabilities from a classification model fit with the real probabilities of the outcomes and how much we can put confidence on predictions of model.

1. Bi-directional Long Short-term Memory



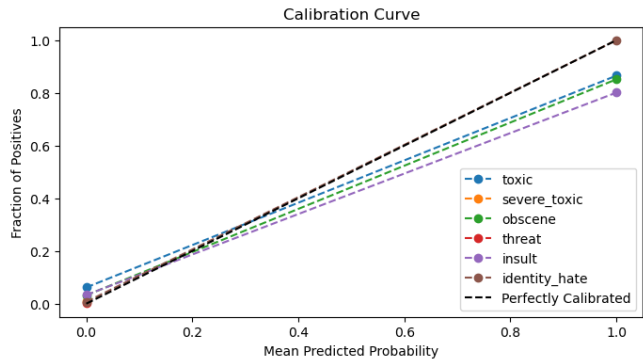
All classes closely followed perfectly calibrated line showing model is perfectly calibrated (Decision thresholds can be set confidently based on predicted probabilities), except few data points on identity hate which is above diagonal line showing class is over-confident (Decision thresholds based on these probabilities might lead to more false positives than expected) in its predictions and severe toxic which is below calibrated line showing class is under-confident (Decision boundaries based on these probabilities may result in missed opportunities (more false negatives).

2. Convolutional Neural Network



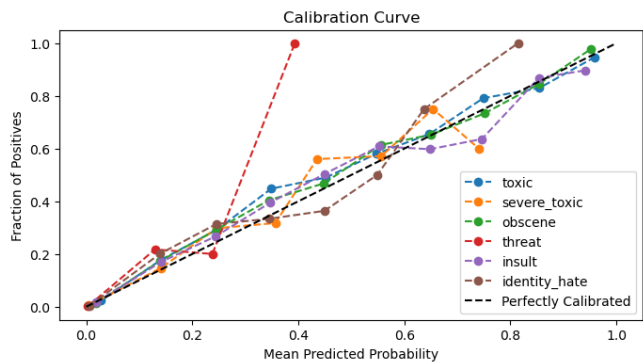
All classes closely followed perfectly calibrated line showing model is perfectly calibrated except few data points of identity hate, severe toxic and obscene show classes are over-confident.

3. Support Vector Machine



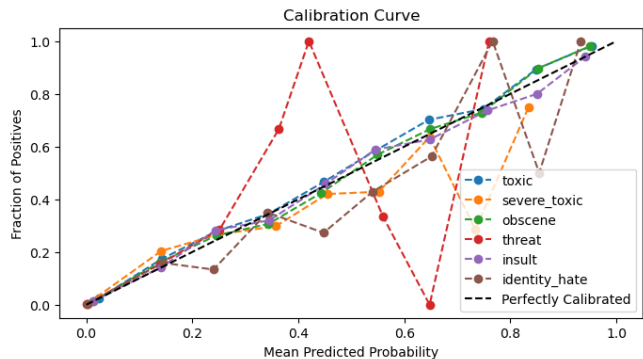
All classes are below perfectly calibrated line showing all classes are under confident.

4. Logistic Regression



Some classes are scattered randomly, not following any pattern or the diagonal line.

5. XGBoost

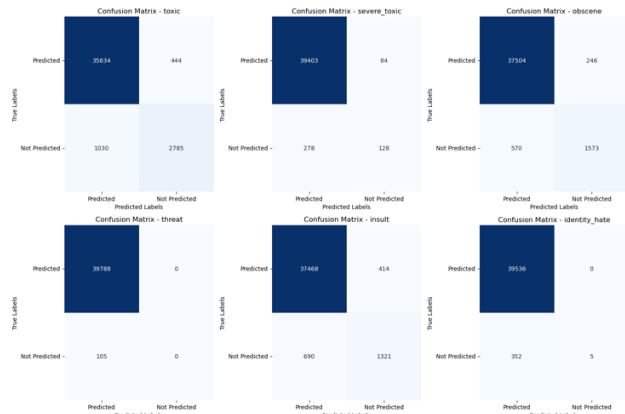


Threat class is fluctuated randomly, identity hate, and severe toxic class is under confidence, while other classes are perfectly calibrated.

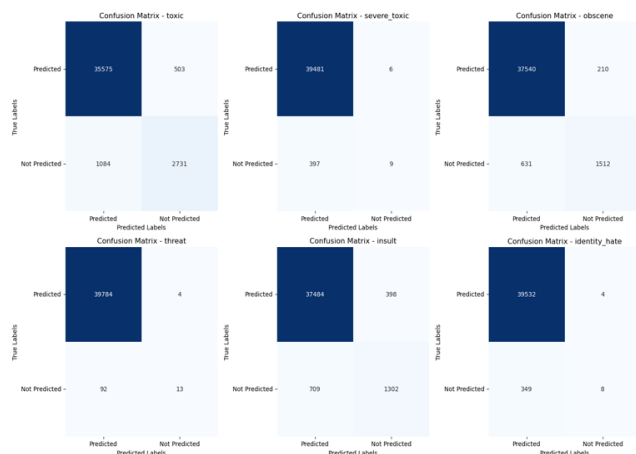
Confusion Matrix

A multi-label confusion matrix is created in the same way as a conventional confusion matrix, except that each member now indicates interactions between true and predicted labels rather than simply true and anticipated classes. It's an effective tool for testing and fine-tuning models in multi-label classification settings.

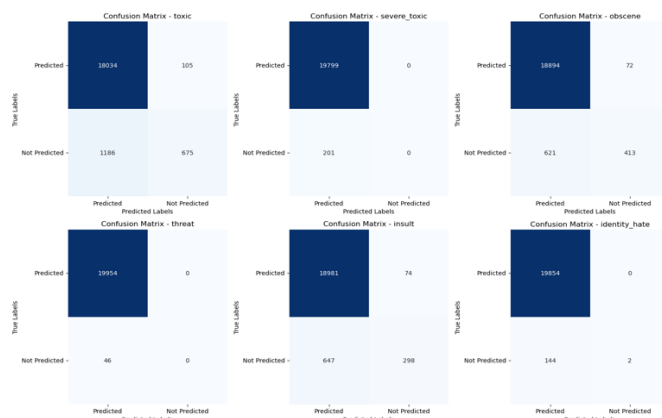
1. Bi-directional Long Short-term Memory



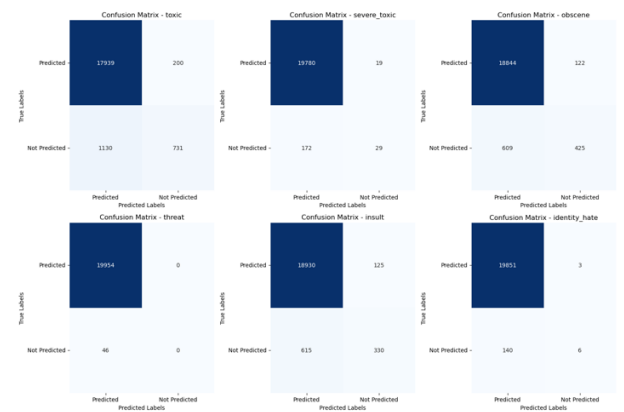
2. Convolutional Neural Network



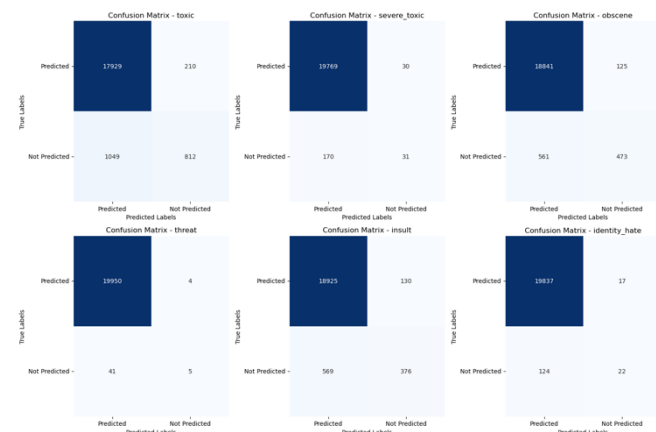
3. Support Vector Machine



4. Logistic Regression



5. XGBoost



DISCUSSION

LSTM is proved effective strategy with combination of word embedding in toxic comments classification because of its ability to capture long sequences in data. Base Classifier such as logistic regression has lowest accuracy. As data is imbalanced so to make sure accuracy did not mislead the result, F1 score metric is used that is effective in such case and penalize the classifier if it is biased toward some classes.

On the other hand, Support Vector Machine has third highest frequency but has lowest F1 score. The reason is that SVM may be biased toward the majority class while doing poorly on the minority class. This can result in a poorer F1 score since the F1 score considers both precision and recall; if the recall for some class is low owing to this imbalance, the F1 score will suffer.

In future, performance of classification models can be increased by comparing performance of different word embedding techniques such as BERT, Word2Vec and Fast Text with different classifiers.

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